SIGNAL INTELLIGENCE DOCUMENT

RigaCap

Signal Intelligence: How RigaCap Reads the Market

A Complete Technical Guide to Our Trading Signal Pipeline — From Raw Data to Portfolio Execution

6,500+	7	5 min	12%
STOCKS SCANNED	MARKET REGIMES	INTRADAY MONITORING	TRAILING STOP

Confidential — February 2026

v1.0.0 · 3-Factor Ensemble · 7-Regime Intelligence

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Executive Summary

RigaCap's signal intelligence pipeline transforms raw market data from 6,500+ US equities into high-conviction trading signals through a three-pillar approach:

DWAP Timing

Proprietary breakout detection from 200-day volume-weighted accumulation zones.

Momentum Quality

Multi-timeframe momentum ranking with volatility-adjusted composite scoring.

Regime Intelligence

7-state market classifier that adapts strategy parameters to current conditions.

Every signal must pass all three filters before entry. Once invested, positions are monitored every 5 minutes during market hours with adaptive trailing stops. The entire pipeline has been validated through 5-year walk-forward testing with zero hindsight bias, producing **+289% cumulative return** (31% annualized).

1. Data Acquisition Pipeline

The pipeline begins with a comprehensive stock universe spanning NASDAQ and NYSE exchanges. Every trading day, the system fetches, filters, and enriches price data to feed the signal engine.

Stock Universe

PARAMETER	VALUE	SOURCE
Total Universe	~6,500 symbols	NASDAQ + NYSE exchanges
Minimum Price	$20	`config.MIN_PRICE`
Minimum Volume	500,000 shares/day	`config.MIN_VOLUME`
Lookback Period	5 years OHLCV	`fetch_data(period="5y")`
Data Source	yfinance (free, no API key)	Split/dividend adjusted

Exclusion Rules

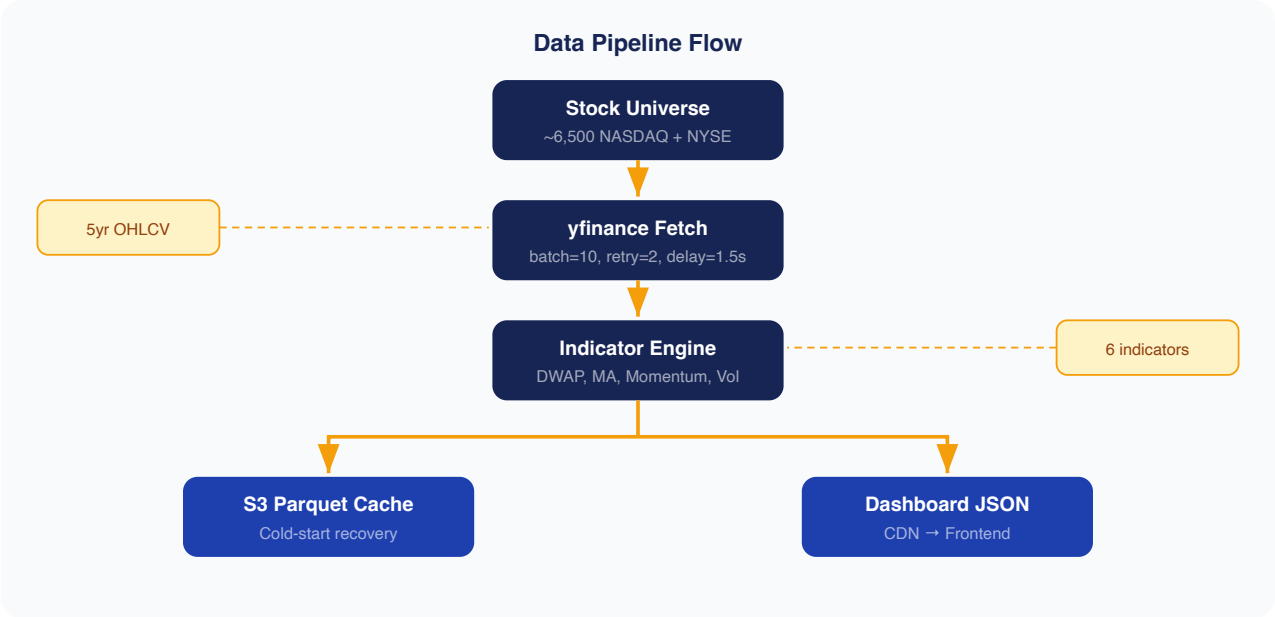
The following instrument types are automatically excluded from the universe:

- **Leveraged ETFs:** TQQQ, SQQQ, SSO, SDS, QLD, QID, TNA, TZA
- **Inverse Products:** SPXU, UVXY, SVXY
- **Volatility Products:** VXX
- **Sector-leveraged:** FAS, FAZ

Fetch Architecture

PARAMETER	VALUE	PURPOSE
Batch Size	10 symbols	Avoid yfinance rate limiting
Max Retries	2 per batch	Handle transient failures
Batch Delay	1.5 seconds	Rate limit compliance
Retry Delay	3.0 seconds (2× batch delay)	Backoff on failure
Required Symbols	SPY, ^VIX	Always fetched for regime detection

After the initial 5-year fetch, subsequent updates use **incremental mode** — only downloading prices since the last cached date. This reduces daily update time from minutes to seconds. All price data is persisted to S3 as consolidated parquet files for Lambda cold-start recovery.

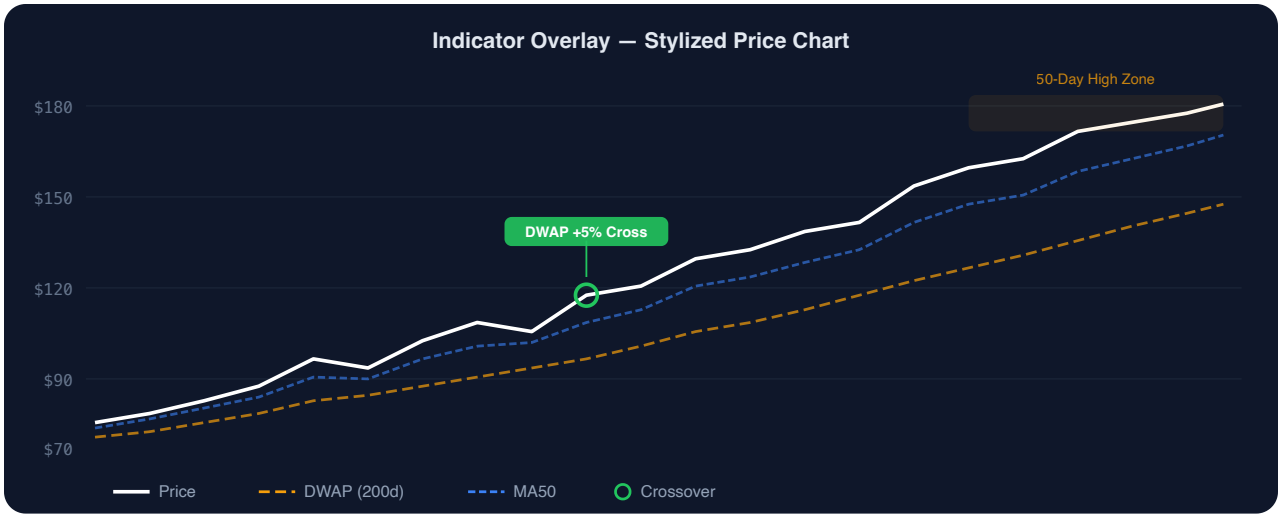


2. Technical Indicator Engine

Six core indicators are computed for every stock in the universe. These indicators feed both the DWAP timing signal and the momentum quality ranking.

INDICATOR	FORMULA	PERIOD	PURPOSE
DWAP	`Σ(price × volume) / Σ(volume)`	200 days	Volume-weighted accumulation zone
MA20, MA50, MA200	Simple Moving Average	20 / 50 / 200 days	Trend confirmation, quality filter
Short Momentum	`(price / price[t−10] − 1) × 100`	10 days	Short-term price acceleration
Long Momentum	`(price / price[t−60] − 1) × 100`	60 days	Sustained trend strength
Volatility	`std(returns, 20) × √252 × 100`	20 days	Risk-adjusted scoring penalty
Distance from High	`(price / max(price, 50d) − 1) × 100`	50 days	Breakout proximity detection

> **Lazy Computation:** Indicators are computed on-demand and cached in the `data_cache` dictionary. The first scan computes all indicators; subsequent scans reuse cached values unless data is refreshed.


Indicator Overlay — Stylized Price Chart
50-Day High Zone
$180
$150
$120
$90
$70
DWAP +5% Cross
Price
DWAP (200d)
MA50
Crossover

3. The Ensemble Signal — Three-Factor Entry

A stock must satisfy all three independent factors simultaneously before the system generates a buy signal. This multi-confirmation approach dramatically reduces false signals and ensures only the highest-conviction setups enter the portfolio.

Factor 1: DWAP Timing

The Daily Weighted Average Price (DWAP) captures the average cost basis of all traders over 200 trading days, weighted by volume. When price breaks above DWAP by 5%, it signals that the stock has cleared its accumulation zone with conviction.

$$\text{DWAP} = \Sigma(\text{price} \times \text{volume})_{200d} \,/\, \Sigma(\text{volume})_{200d}$$

$$\text{BUY when price} > \text{DWAP} \times 1.05$$

Factor 2: Momentum Quality

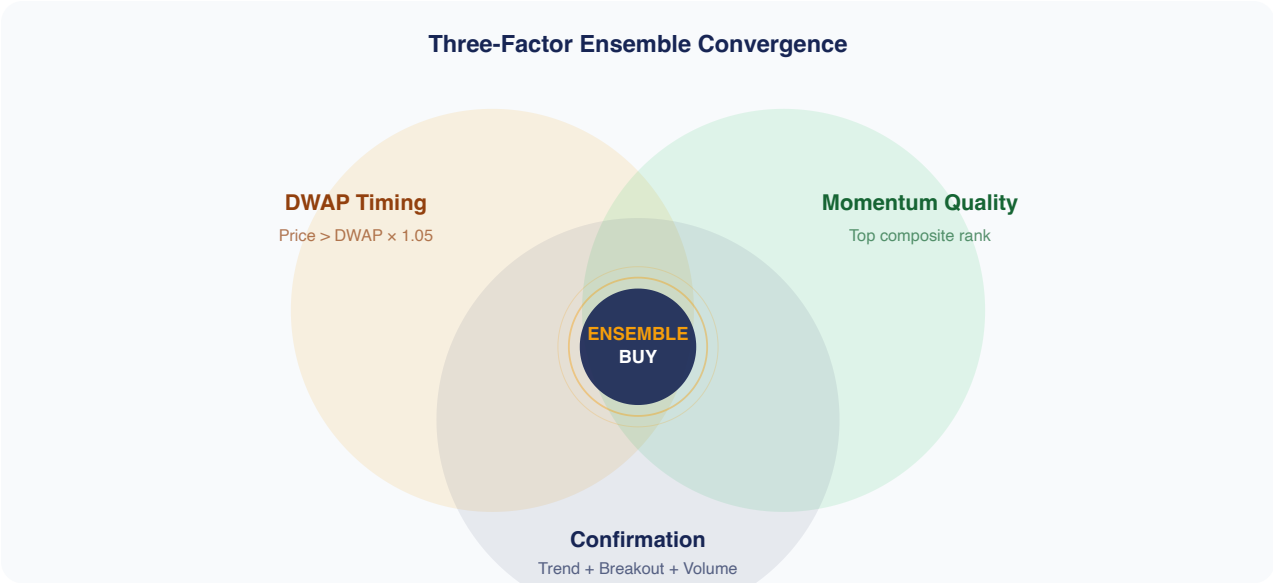
Stocks passing the DWAP filter are ranked by a composite momentum score that rewards short-term acceleration and sustained strength, while penalizing excessive volatility:

$$\text{composite} = \text{short_mom}_{10d} \times 0.5 + \text{long_mom}_{60d} \times 0.3 - \text{volatility}_{20d} \times 0.2$$

Factor 3: Confirmation Filters

FILTER	CONDITION	PURPOSE
Uptrend	Price > MA20 AND Price > MA50	Confirms healthy trend structure
Breakout	Within 5% of 50-day high	Near breakout zone, not exhausted
Volume	≥ 500,000 shares/day	Sufficient liquidity
Price	≥ $20	Filters penny stocks
Sector Cap	Max 5 per sector	Prevents sector concentration

Strong Signal Bonus: Stocks with volume ≥ 1.5× their 200-day average AND price above both MA50 and MA200 (healthy trend) are marked as "strong" signals — indicating institutional participation.



4. Market Regime Intelligence

The market regime classifier analyzes SPY price action, VIX levels, market breadth, and new-highs data to classify the current environment into one of 7 distinct regimes. Strategy parameters automatically adapt to each regime.

7 Regime Classifications

Strong Bull
SPY +5% above 200MA
VIX < 40th percentile

Weak Bull
SPY 0–10% above 200MA
Moderate volatility

Rotating Bull
SPY choppy, 1yr gain +5%
Sector rotation

Range-Bound
SPY ±5% of 200MA
Flat 1-year return

Weak Bear
SPY −15 to 0% vs 200MA
Elevated VIX

Panic / Crash
SPY < −10% vs 200MA
VIX > 70th percentile

Recovery
Bouncing from lows
SPY 0–15% above 50MA

+ Forecasting
1–2 week transition probability outlook

Scoring Weights

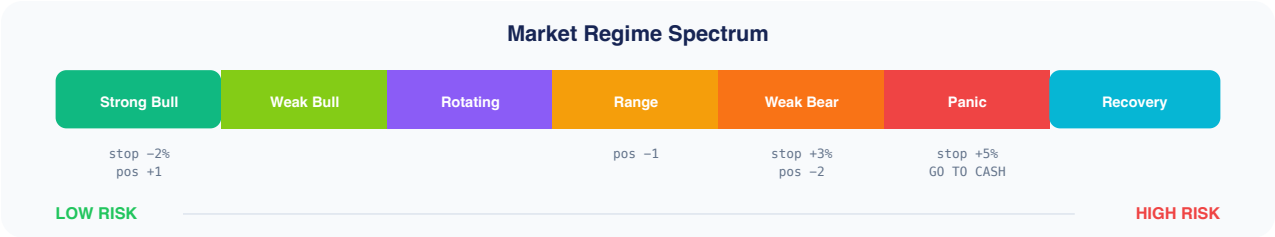
Regime detection uses weighted scoring across condition indicators. Higher weights mean that indicator is more influential in regime classification:

INDICATOR	WEIGHT	ROLE
Long-term Trend (252d SPY return)	2.5×	Key differentiator: rotating bull vs range-bound
SPY vs 200MA	1.5×	Primary bull/bear boundary
New Highs %	1.5×	Breadth confirmation of trend
Breadth % (above 50MA)	1.2×	Market participation width
VIX Percentile, Trend Strength, SPY vs 50MA	1.0×	Standard conditions

Regime Parameter Adjustments

REGIME	TRAILING STOP	MAX POSITIONS	POSITION SIZE	ACTION
Strong Bull	−2% (tighter)	+1	+2%	Stay invested

REGIME	TRAILING STOP	MAX POSITIONS	POSITION SIZE	ACTION
Weak Bull	No change	No change	No change	Stay invested
Rotating Bull	+1% (wider)	No change	No change	Stay invested
Range-Bound	No change	−1	No change	Tighten stops
Weak Bear	+3% (wider)	−2	−3%	Reduce exposure
Panic/Crash	+5% (widest)	−3	−5%	Go to cash
Recovery	+2%	No change	No change	Gradual re-entry



Transition Forecasting

Beyond static classification, the regime engine computes **indicator trajectories** to predict where the market is heading in the next 1–2 weeks:

- **VIX Delta (5-day):** Rising VIX boosts bearish regime probabilities
- **Trend Acceleration:** Compares recent 5-day return vs prior 5-day return
- **200MA Proximity:** Within 2% of 200MA triggers higher transition uncertainty
- **Sharp Moves:** 5-day returns exceeding ±5% boost crash/recovery probabilities

The forecast produces a **recommended action** ranging from `stay_invested` → `tighten_stops` → `reduce_exposure` → `go_to_cash`.

5. Intraday Monitoring (Every 5 Minutes)

Once positions are entered, the intraday monitor checks live prices every 5 minutes during market hours. This enables rapid response to trailing stop hits and regime-driven exits.

Schedule

PARAMETER	VALUE
Hours	Monday–Friday, 9:00–15:55 ET
Interval	Every 5 minutes
Price Source	`yfinance fast_info.last_price`
Alert Deduplication	`(position_id, action, date)`

Trailing Stop Mechanics

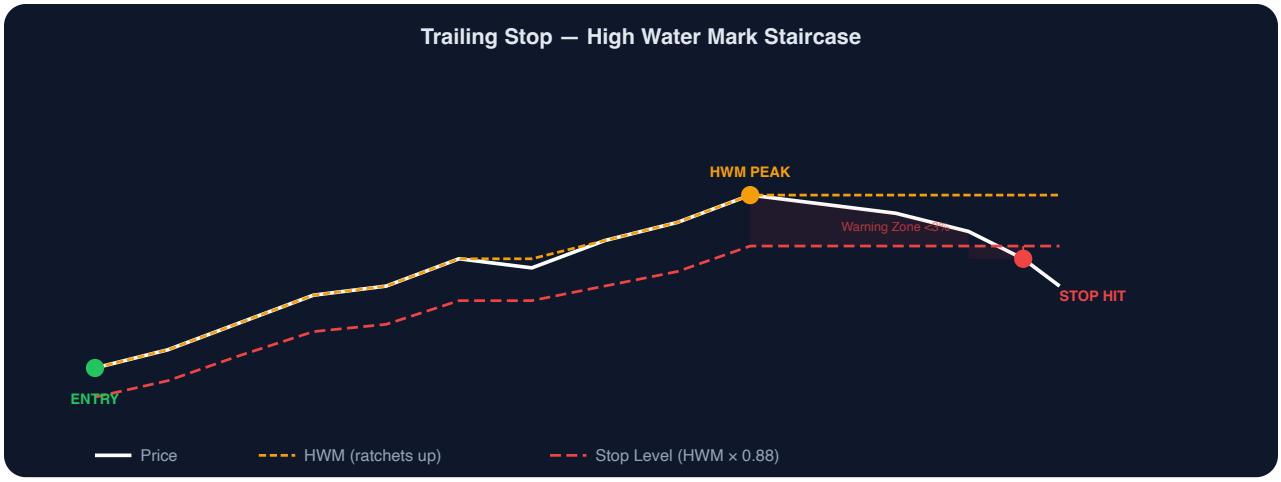
The trailing stop is calculated from the **high water mark** (HWM) — the highest price since entry, not the entry price itself:

```
HWM = max(entry_price, stored_highest, current_price)
stop_level = HWM × (1 − 0.12)

SELL when price ≤ stop_level     WARNING when distance < 3%
```

Exit Priority

1. **Regime Exit (go_to_cash):** Immediate sell regardless of P&L — capital preservation
2. **Trailing Stop Hit:** Price ≤ HWM × 0.88 — overrides regime warnings
3. **Warning Zone:** Within 3% of stop level — email alert, no action
4. **Regime Warning:** reduce_exposure or tighten_stops — email alert



Intraday DWAP Crossover Detection

The monitor also watches stocks on the **dashboard watchlist** (stocks approaching the DWAP threshold). If a watchlist stock crosses DWAP +5% intraday, the system:

- Adds the stock to the dashboard as a fresh intraday signal
- Removes it from the watchlist
- Sends intraday signal alert emails to subscribed users (skipping users who already hold the stock)
- Updates the S3 dashboard JSON in real-time

6. Position Management & Sizing

The model portfolio service maintains two parallel portfolios — live (intraday monitoring) and walk-forward (biweekly rebalancing) — plus three ghost portfolios for comparison.

Portfolio Parameters

6	15%	12%	14d
MAX POSITIONS	POSITION SIZE	TRAILING STOP	REBALANCE PERIOD

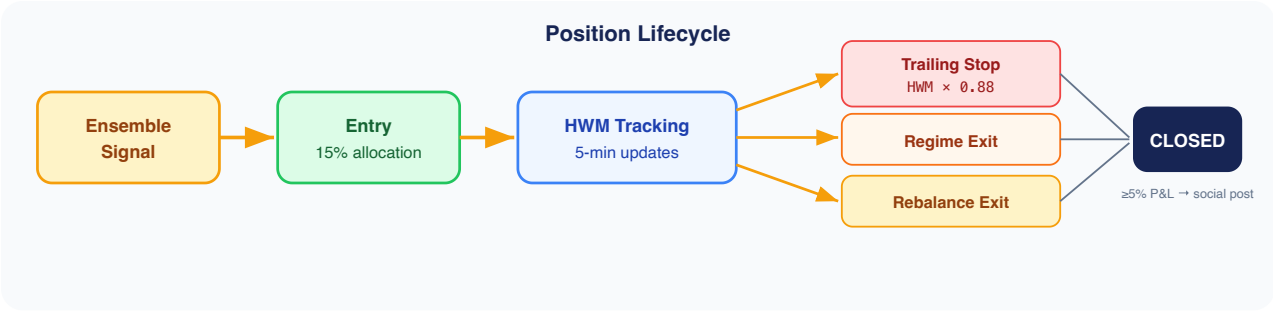
Entry Logic

- Read fresh ensemble signals from dashboard cache (S3 JSON)
- Filter to `is_fresh = true` only, skip symbols already held
- Sort by ensemble_score (highest first)
- Enter up to `MAX_POSITIONS - open_count` new positions
- Allocate 15% of **current cash** per position (not starting capital)
- Minimum allocation: $100

Exit Paths

EXIT REASON	TRIGGER	PORTFOLIO
`trailing_stop`	Price ≤ HWM × 0.88	Both live & WF
`regime_exit`	`go_to_cash` recommendation	Live only
`rebalance_exit`	Biweekly period boundary	Walk-forward only

> **Biweekly Boundaries:** Walk-forward periods are 14-day intervals anchored to February 1, 2026 (`WF_ANCHOR_DATE`). At each boundary, all open WF positions are force-closed and fresh signals are entered for the new period.



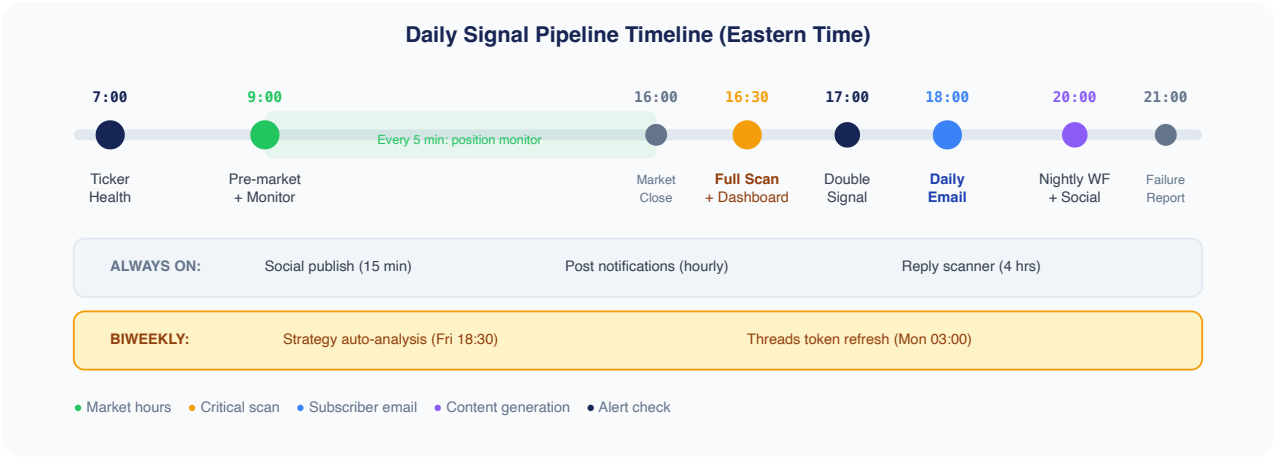
Ghost Portfolios (Parallel Universes)

Three alternative portfolios run alongside with different parameters to prove the chosen configuration:

GHOST	TRAILING STOP	MAX POSITIONS	POSITION SIZE	STYLE
Aggressive	8%	8	12%	Tight stops, more positions
Conservative	18%	4	20%	Wide stops, fewer positions
Top-3 Only	12%	3	30%	Concentrated best picks

7. Alert & Communication Pipeline

A coordinated set of scheduled jobs runs throughout each trading day, from pre-market data refresh through nightly content generation. Every job is managed by APScheduler with US Eastern timezone awareness.



Alert Types

ALERT	TRIGGER	AUDIENCE	CHANNEL
Daily Digest	6 PM ET, trading days	All subscribers	Email + Push
Sell Alert	Trailing stop or regime exit	Position holders	Email + Push
Warning Alert	Within 3% of stop level	Position holders	Email + Push
Double Signal	DWAP + momentum crossover within 3 days	All subscribers	Email
Intraday Signal	Watchlist stock crosses DWAP +5%	Subscribers (not holding)	Email
Ticker Health	Data issues detected at 7 AM	Admin only	Email

Email Deliverability: All emails are sent with SPF + DKIM + DMARC DNS records configured, plus RFC 8058 List-Unsubscribe + List-Unsubscribe-Post headers for one-click unsubscription. Each email includes a user-specific unsubscribe link for per-category preference control.

8. Walk-Forward Validation

Walk-forward testing is the gold standard for trading system validation. Unlike traditional backtesting which optimizes on all available data (introducing hindsight bias), walk-forward divides history into sequential periods and tests forward into data the optimizer has never seen.

Methodology

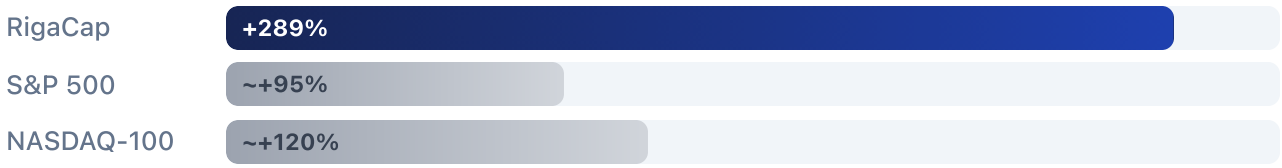
- **Period Length:** Biweekly (14-day intervals, anchored Feb 1)
- **Optimization:** Uses only historical data up to each period boundary
- **Trading:** Enters fresh ensemble signals, force-closes at period end (`rebalance_exit`)
- **Position Carry:** Force-close outperforms carry-over (A/B tested)
- **Max Symbols:** 100 per period for computational efficiency
- **Strategy:** Ensemble (strategy_id: 5) — DWAP + Momentum + Regime

> **Zero Hindsight Bias:** At each biweekly boundary, the system knows nothing about what happens next. Parameters are optimized on past data, then the system trades forward into unseen market conditions. The 289% return is entirely out-of-sample.

5-Year Results (2021–2026)

PERIOD	RETURN	SHARPE	MAX DRAWDOWN	MARKET CONTEXT
2021–2022	+62.0%	1.21	-14.8%	Post-COVID rally, meme mania
2022–2023	-13.2%	-1.38	-15.1%	Fed rate hikes, tech selloff
2023–2024	+22.2%	1.02	-13.6%	AI-driven recovery
2024–2025	+20.7%	0.89	-13.7%	Election volatility, rate cuts
2025–2026	+87.5%	2.32	-8.3%	Strong bull, tariff recovery
5-Year Total	**+289%**	**1.02 median**	**-15.1%**	**31% annualized**

RigaCap vs. Benchmarks (5-Year Cumulative)



Key Insights

80% Win Rate

4 of 5 years profitable. The single losing year (2022-2023, -13.2%) was limited by regime detection while the S&P 500 fell ~20%.

Controlled Drawdowns

Maximum drawdown of -15.1% across all market conditions — the trailing stop and regime system prevented catastrophic losses.

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